

EDGE-CASE TEST MATRIX: NORTHSTAR SUPPORT CHATBOT

Project: Northstar Support Chatbot

1. Execution Context & Verification Scope

This document details the comprehensive edge-case testing executed for the **Northstar Support Chatbot**. Each scenario validates frontend formatting rules, business logic enforcement (30-day window, non-returnable policies, damage fast-track), and error handling with Supabase integration.

- **Local Verification Protocol:** Execute `npm run dev` in local terminal, open browser at `http://localhost:5173/`, and input test strings sequentially into chat interface.

2. System Edge-Case Test Execution Matrix

Test ID	Scenario	Input	Expected Behavior	Actual Result	Status
TC-01	Valid Eligible Order	ORD1001	Order details retrieved from Supabase; return options displayed for eligible item(s).	Returns order details & eligible options	PASS
TC-02	Expired Return Window	ORD1002	Order retrieved; return request blocked due to exceeding the 30-day policy window.	System informs user order is outside 30-day window	PASS
TC-03	Final Sale / Non-Returnable	ORD1003	Order retrieved; return denied based on final_sale / non-returnable policy tag.	System flags item as non-returnable	PASS
TC-04	Damaged Goods Fast-Track	ORD1004	Order retrieved; system skips auto-approval and fast-tracks escalation to human support.	Prompt triggers agent escalation flow	PASS
TC-05	Missing ID Prefix	1001	Validation error triggered; prompts user to format order ID properly (e.g., ORDXXXX).	Prompts user for ORD prefix format	PASS
TC-06	Non-Existent Order ID	ORD9999	Query executes successfully but returns zero matches; displays 'Order not found' error.	Shows 'Order not found' message	PASS
TC-07	Missing .env Credentials	Unset .env	Supabase client fails to initialize gracefully; displays 'Error connecting to database service'.	Displays database error banner	PASS

QA Sign-off Status: 7 / 7 TESTS PASSED (100% PASS RATE) |